RANJAN KUMAR

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2026	PhD.	*IIT Kharagpur (About to complete in 2 month)*	8.77 / 10
2019	M. Tech	*CAET, Godhra Anand Agricultural University (AAU)*	8.78 / 10
2017	B. Tech	*KCAEFT, Thavanur, Kerala Agricultural University (KAU)*	7.70 / 10
2012	AISSCE	*Central Board of Secondary Education*	69.6%
2010	AISSE	*Central Board of Secondary Education*	8.0 / 10

INTERNSHIPS/EXPERIENCES

- Served as a **Teaching Assistant** at IIT Kharagpur, which, included Subject as Engineering Drawing (CE13001) during the 2020–22 spring semesters, alongside Irrigation and Drainage Engineering Lab (AG39004) and Geo-informatics for Land and Water Resources Lab (AG69039) across consecutive spring and autumn semesters from 2022 to 2025.
- **Remote Sensing & Water Resources**: Completed an NNRMS-sponsored internship at IIRS (2015), along with specialized training in groundwater investigation and soil/water conservation with Kerala government departments (2017).
- **Agricultural & Food Processing:** Undertook a series of brief internships in spring 2017 across IISR, SRFMTTI, KVASU Dairy Plant, and Pavizham Healthier Diet Pvt. Ltd. in Kerala.
- **Professional Development**: Completed a two-day training course on physical and numerical modeling for hydraulic reservoir design.
- Run **uTrack moisture model** on Param Shakti High-Performance Computing system at IIT Kharagpur for Physics-based Climate Runs.

SKILLS AND EXPERTISE

Programming Languages: C | Python | MATLAB |NCL

Libraries: Pandas | NumPy | Matplotlib | NetCDF | Xarray

Software and Skills: Bash Scripting | LINUX | Spyder | GitHub | Remote Sensing | ARCGIS | ERDAS Imagine | Google Earth Engine | AutoCAD | Big Climatic data Analysis |CDO |NCO |Eulerian Lagrangian moisture tracking model (Utrack)

PUBLICATIONS

- **Kumar, R.,** Pathak, A., & Kulkarni, S. (2026). Climatic and transient controls on Indian monsoon EP variability using moisture-budget decomposition**. Environmental Research Communications**.
- **Kumar, R.,** & Pathak, A. (2025). Lagrangian quantification of atmospheric moisture sources for extreme rainfall events over India during the 2023 summer monsoon. **Science of The Total Environment**, 1000, 180389.
- **Kumar, R.,** Ghosh, S., Patel, G., Singh, L. K., Mohanty, A. K., & Singha, A. K. (2025). Mapping Wheat Potential: A GIS-Based Agro-Land Suitability Analysis in Gujarat's Bhal Region, Western India. **Frontiers in Agronomy**, 7, 1551494
- **Kumar, R.,** Joseph, A., & Poltharan, A. (2026). Experimental Investigation on Sustainable Soil Stabilization Using Coconut Coir Fibre**. Journal of Scientific Research and Reports**, 32(3), 482-500.
- **Kumar, R**. and Patel, G.R., 2020. Assessment of Agro-Land Suitability for Rice (Oryza sativa L.) in Bhal Area of Gujarat Using GIS and Remote Sensing**. International Journal of Current Microbiology and Applied Sciences**, 9(4), pp.1207-1214

CONFERENCES

- **Kumar, R.,** & Pathak, A. (2022, December). Diagnostic Evaluation of Atmospheric Moisture Budget (EP) during ENSO years and its relation to ISM rainfall variability. In **AGU Fall Meeting Abstracts** (Vol. 2022, pp. H44C-07).
- **Kumar, R.,** Pramod, & Pathak, A. (2024, June). Detection of atmospheric rivers associated with extreme precipitation over the Indian subcontinent. In **International Atmospheric Rivers Conference 2024**, June 24–27, 2024 (Vol. 5, Issue 67, pp. 73–96). Scripps Institution of Oceanography, La Jolla. Book of Abstracts.
- **Kumar, R.,** & Pathak, A. (2025, April). Tracing Moisture Flow from the Bay of Bengal and Arabian Sea with its Impact on Ganga Basin during Monsoonal Extremes**.** In **EGU General Assembly Conference** Abstracts (pp. EGU25-4531).
- **Kumar, R.,** Chouhan, P., & Pathak, A. (2024, December). Identification of Atmospheric Rivers Linked to Extreme Precipitation in the Core Monsoon Zone of India. In **AGU Fall Meeting Abstracts** (Vol. 2024, pp. A07-03).
- **Kumar, R.,** & Pathak, A. (2024, December). Role of Western Indian Ocean on the Indian Monsoon Extreme Events during 2023. In **AGU Fall Meeting Abstracts** (Vol. 2024, pp. A43Y-03).
- **Kumar, R.,** & Pathak, A. (2025, October). *Atmospheric moisture budget and its variability over the Indian subcontinent*. In **XIIth Scientific Assembly of the International Association of Hydrological Sciences (IAHS 2025).** IIT Roorkee, Roorkee, Uttarakhand, India.
- **Kumar, R.,** & Pathak, A. (2026). Chasing the sources of India's extreme rainfall: Insights from UTrack and HYSPLIT. Presented at the **Roorkee Water Conclave 2025–26**, IIT Roorkee, Roorkee, Uttarakhand, India.

AWARDS AND ACHIEVEMENTS

- **National Examinations:** Qualified the ASRB-ICAR-NET (2020) with 57.56%; secured AIR-36 in ICAR-AICE (PhD, 2019), AIR-42 in ICAR-AIEEA (PG, 2017), and qualified JEE Mains (2013) & eligible to appear in JEE Advanced.
- **Academic Excellence**: Graduated as the **M.Tech College Topper** among **18 students** across all departments at Anand Agricultural University, and Selected for **Ph.D** among approximately **30** applicants in the competitive IIT Kharagpur PhD entrance examination.
- **Leadership & Sports**: Awarded the prestigious Pt. Ishwar Chandra Vidyasagar Award (2024) for outstanding hall welfare contributions at IIT Kharagpur, alongside winning consecutive Gold Medals in the Interhall Badminton Tournaments (2022–2023).

REFERENCES

- **Dr. Amey Pathak**, Assistant Professor, Department of Agricultural and Food Engineering, Indian Institute of Technology Kharagpur, Kharagpur, West Bengal, India – 721302, Email: pathak.amey1@gmail.com.